```
47
48      // 用直线段连接相邻的采样点，绘制曲线
49      for (size_t i = 0; i < curvePoints.size() - 1; i++) {
50          pRT->DrawLine(curvePoints[i], curvePoints[i + 1], pBrush, 1.0
                f);
51      }
52  }
```

Listing 8: de Casteljau 算法实现

5.5 主程序框架

```
1  /**
2   * 创建Direct2D设备相关资源
3   * @param hWnd 窗口句柄
4   * @return 操作结果
5   */
6  HRESULT CreateDeviceResources(HWND hWnd) {
7      // 获取窗口客户区大小
8      RECT rc;
9      GetClientRect(hWnd, &rc);
10     D2D1_SIZE_U size = D2D1::SizeU(rc.right, rc.bottom);
11
12     // 创建硬件渲染目标
13     HRESULT hr = g_pD2DFactory->CreateHwndRenderTarget(
14         D2D1::RenderTargetProperties(),
15         D2D1::HwndRenderTargetProperties(hWnd, size),
16         &g_pRenderTarget);
17
18     // 创建画刷
19     if (SUCCEEDED(hr)) {
20         hr = g_pRenderTarget->CreateSolidColorBrush(
21             D2D1::ColorF(D2D1::ColorF::Black), &g_pBrush);
22     }
23     return hr;
24 }
```

Listing 9: Direct2D 初始化

```
1  /**
2   * 处理WM_PAINT消息，绘制所有图形
```